

Unit 8.6 The Fundamental Theorem of Calculus.

Note Title

5/14/2018

If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a),$$

where F is any anti-derivative of f , that is:
 $F'(x) = f(x)$.

Proof: See study guide (p. 27) for the complete proof.

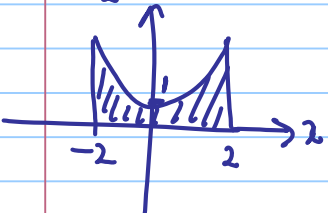
This proof is also discussed in a video.

Examples:

1) $\int_0^{\pi/2} \sin x dx = -\cos x \Big|_0^{\pi/2}$ $\frac{d}{dx}(-\cos x) = \sin x$


$$\begin{aligned} &= -\cos \frac{\pi}{2} - (-\cos 0) \\ &= -0 + \cos 0 \\ &= -0 + 1 \\ &= 1. \end{aligned}$$

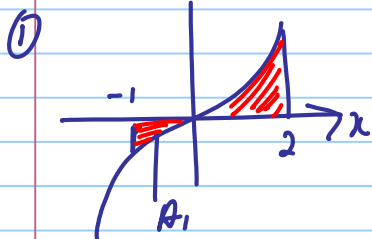
2) $\int_{-2}^2 (x^2 + 1) dx = \left[\frac{1}{3} x^3 + x \right]_{-2}^2$


$$\begin{aligned} &= \frac{1}{3}(8) + 2 - \left(\frac{1}{3}(-8) - 2 \right) \\ &= \frac{8}{3} + 2 + \frac{8}{3} + 2 \\ &= \frac{16}{3} + 4. \end{aligned}$$

Area problems:

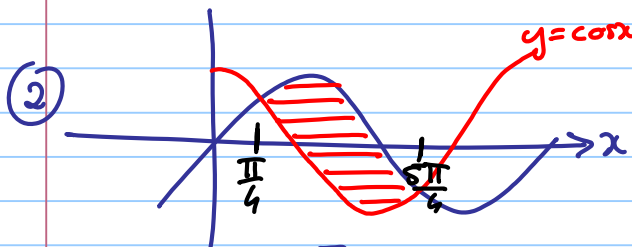
- ① Find the area enclosed between $y = x^3$ and the x -axis between $x = -1$ and $x = 2$.
- ② Find the area enclosed between $y = \sin x$ and $y = \cos x$, between $x = \pi/4$ and $x = 5\pi/4$.
- ③ Find the area enclosed between $y = x^2$ and $y = x$ between $x = 0$ and $x = 2$.

Solutions of area problems: Always sketch (if possible)
MB



Note: $\int_{-1}^0 x^3 dx$ is negative
If we want the area of A_1 , then it is $-\int_{-1}^0 x^3 dx$

$$\begin{aligned} \text{Area} &= -\int_{-1}^0 x^3 dx + \int_0^2 x^3 dx \\ &= -\left[\frac{1}{4}x^4\right]_{-1}^0 + \left[\frac{1}{4}x^4\right]_0^2 \\ &= -\left[0 - \frac{1}{4}(-1)^4\right] + \left[\frac{1}{4}(2^4) - 0\right] \\ &= -\left[-\frac{1}{4}\right] + \frac{16}{4} \\ &= \frac{1}{4} + \frac{16}{4} \\ &= \frac{17}{4} \end{aligned}$$



$$\begin{aligned} \text{Area} &= \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx \\ &= \left[-\cos x - \sin x\right]_{\pi/4}^{5\pi/4} \end{aligned}$$

$$\begin{aligned} &= -\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} - \left[-\cos \frac{\pi}{4} - \sin \frac{\pi}{4}\right] \\ &= -\left(-\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \end{aligned}$$

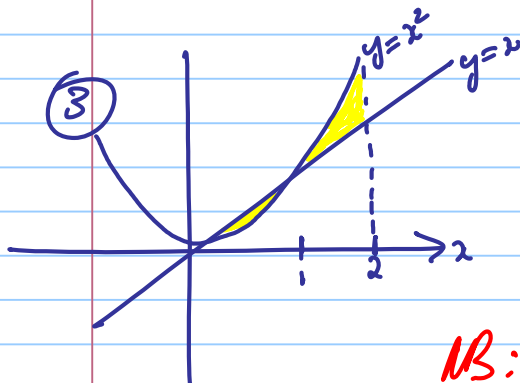
$$\cos \frac{5\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$\sin \frac{5\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$= \frac{4}{\sqrt{2}}$$

$$= 2\sqrt{2}$$



MB: Sketch!!

$$\begin{aligned} \text{Area} &= \int_0^1 (x - x^2) dx + \int_1^2 (x^2 - x) dx \\ &= \end{aligned}$$

MB: Make sure you have this correct!!